

Nepal Engineering College

Digital Signal Analysis & Processing

Tutorial 3

Chapter 3: Analysis of LTI System in frequency domain

1. Plot magnitude and phase response of the following system

a. $H(z) = \frac{1-z^{-1}}{1-0.5z^{-1}}$

b. $H(z) = \frac{1-2z^{-1}}{1-0.6z^{-1}+0.2z^{-2}}$

2. Determine the output sequence of the system with impulse response

$$h[n] = 0.5^n u[n]$$

when the input is the complex exponential sequence

$$x[n] = Ae^{\frac{j\pi n}{2}}, -\infty < n < \infty$$

Chapter 4: Discrete-Filter Structure

1. Draw direct form-I and form-II structure of

$$y[n] = -0.9y[n-1] + 0.6y[n-2] - 0.1y[n-3] + 3x[n] + 2x[n-3]$$

2. Draw cascade and parallel structure

$$H(z) = \frac{1 + 2z^{-1} + 3z^{-2}}{1 - 0.9z^{-1} + 0.6z^{-2} + 0.2z^{-3}}$$

3. Draw lattice ladder structure of

a. $\frac{1+2z^{-1}+3z^{-2}}{1-0.9z^{-1}+0.6z^{-2}}$

b. $\frac{1+2z^{-1}+3z^{-2}+z^{-3}}{1-0.9z^{-1}+0.6z^{-2}+0.2z^{-3}}$

Tutorial 4:

Chapter 7: Discrete Fourier Transform

1. Determine the 4-point DFT of the signal $x[n] = u[n] - u[n-3]$
2. Computer the circular convolution of the following sequences
 $x[n] = \{0, 1, 2, 3, 1\}$ & $h[n] = \{1, 1, 2, 1\}$
3. Use the butterfly diagram to compute 8-point DIT-FFT of the following sequence using radix-2 decimation in time algorithm,
 $X[n] = \{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\}$
4. Use the butterfly diagram to compute 8-point FFT of the following sequence using radix-2 decimation in frequency algorithm,
 $X[n] = \{1, 1, 0, 0, -1, -1, 0, 0\}$
5. A FIR filter with impulse response $h[n] = \{1, 2, 4, 3\}$ is excited with the input sequence $x[n] = \{1, 2, -1\}$. Use circular convolution using circular graphs to compute the output of the filter.

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Tutorial 5:

Chapter 6: FIR Filter Design

1. Design a lowpass filter which will have -3dB cut off at 30π rad/sec and an attenuation of 50 dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec.
2. A low pass filter is desired to have frequency response $H_d(\omega)$ as defined below. Design FIR filter using Blackman window to meet the following requirement, the cut-off frequency $\omega_c = \pi/3$ and order $N = 5$. Does this filter have linear phase?

$$H_d(\omega) = \begin{cases} e^{-\frac{j(N-1)}{2}\omega}; & 0 \leq |\omega_c| \\ 0; & \text{otherwise} \end{cases}$$

The Blackman window is defined as,

$$W[n] = \begin{cases} 0.42 - 0.5 \cos \frac{2\pi n}{N-1} + 0.08 \cos \frac{4\pi n}{N-1}, & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

3. Design an FIR linear phase filter using Kaiser Window to meet the following specifications.

$$\begin{aligned} 0.99 \leq |H(e^{jw})| \leq 1.01, & \text{ for } 0 \leq |w| \leq 0.19\pi \\ |H(e^{jw})| \leq 0.01, & \text{ for } 0.21\pi \leq |w| \leq \pi \end{aligned}$$

4. Explain the followings
 - a. FIR filter using rectangular window
 - b. Gibbs phenomenon
 - c. Linear phase filter
 - d. Comparison of different window function

Chapter 7: IIR Filter design

1. Derive a transfer relation of IIR filter between s-plane and z-plane by using Impulse invariance and Bilinear transformation method.
2. Convert the analog filter into a digital filter whose system function is

$$\begin{aligned} \text{a. } H(s) &= \frac{s+0.2}{(s+0.2)^2+9} \\ \text{b. } H(s) &= \frac{1}{(s+0.5)(s^2+0.5s+2)} \end{aligned}$$

Use impulse invariant technique. Assume $T = 0.1$ sec.

3. Convert the analog filter with system function

$$H(s) = \frac{s+0.1}{(s+0.1)^2+9}, \omega_r = \frac{\pi}{4}$$

Into a digital IIR filter using Bilinear Transformation.

4. Design a lowpass Butterworth filter to meet the following specifications.
Passband gain = 0.89 Passband frequency edge = 30 Hz
Attenuation = 0.20 Stopband edge = 75 Hz